When does modelling the intermediate help?

Composite versus Direct, with both surrogates fitting their own hyperparameters

Summary

Every problem here is $f(x) = g(h(x))$ with h expensive and unknown and g cheap and known. Two surrogates compete: **Direct** puts a GP on f; **Composite** puts a GP on h and pushes it through g.

Three findings, in order of how much they change the picture.

1. **Composite's advantage on bijective maps was an artefact of the setup, not a property of the problem.** An earlier version of this document reported that Composite wins almost everywhere, bijective or not. In that grid Composite was handed the generator's true kernel while Direct had to estimate one, and h was drawn from an RBF kernel while both surrogates assumed Matérn. Remove both advantages and the bijective column collapses: five of six bijective maps now show no effect at all. On `identity` the gain fell from $+0.409$ to $+0.087 \pm 0.155$.
2. **With that fixed, bijectivity does predict the gain.** All ten non-bijective maps gain, every one significant at 0.05, by up to 3.4 orders of magnitude. Five of six bijective maps do not.
3. **But information is not the mechanism.** The number of preimages $\bar{n}$ has essentially no correlation with the size of the gain ($r = -0.12$). What does predict it is how *flat g is at its own optimum* ($r = +0.78$). The two come apart cleanly on `shiftsq_c3`, which is bijective in practice and still gains $+2.36$, and on `sum`, which destroys information and gains nothing.

The rest of the document gives the setup, the problems, the numbers and the evidence for each claim.

1 The setup

$$f(x) \;=\; g\big(h(x)\big), \qquad x \in [0,1]^6,$$

with h the *intermediate* (expensive, unknown) and g the *outer map* (cheap, known in closed form).

	Observes	Models	Predicts f by
Direct	$f(x_i)$ only	a GP on f	reading off that GP
Composite	$h(x_i)$	a GP on each h_j	sampling h and applying g

Both get the same inputs, the same budget of 45 evaluations, the same initial design and the same acquisition function, and **both fit their own kernel hyperparameters by maximum likelihood**. That last point is what changed. Only the surrogate differs.

1.1 The intermediates

h is not an analytic test function. It is a draw from the *same* Gaussian process family the surrogates assume, built from random Fourier features:

$$h(x) = \sqrt{\tfrac{2}{M}} \sum_{m=1}^{M} a_m \cos(\omega_m^\top x + b_m), \qquad a_m \sim \mathcal{N}(0,1),\ b_m \sim U(0, 2\pi),$$

with $M = 1024$ and $\ell = 0.5$, then standardised to mean 0 and standard deviation 1.

The frequencies ω_m are drawn from the kernel's spectral density. For RBF that is Gaussian; for Matérn-ν it is a multivariate Student-t with 2ν degrees of freedom, which is what makes Matérn draws rougher. **We use Matérn-5/2, because that is BoTorch's default kernel.** With an RBF draw the surrogates are merely close to correct about h; with a Matérn draw they are exactly correct, and the comparison is no longer confounded by a mismatch that happens to hurt one arm more than the other.

Drawing h from the surrogate's own prior is the point of the design. If h were Rastrigin, Composite might win or lose because a GP models *that function* well or badly, and we would learn nothing about g. Here Composite's model of h is correct by construction, so every remaining difference is attributable to the outer map. Standardising puts the mean at zero so h takes both signs; without that, h^2 would be injective over the visited region and the question would be empty.

Range. Over $[0,1]^6$, h reaches roughly $[-4.3, +3.8]$, with 5% of the domain beyond $|h| > 2$ and 0.3% beyond $|h| > 3$. The range saturates — sampling a million points finds no more than ten thousand — because it is a property of the function, not the sampling. That is why a root at ± 2 sits inside the bulk while a vertex at $h = 3$ sits at the edge of what is reachable, which turns out to matter a great deal.

Several intermediates. When $k > 1$ each h_j is an independent draw with its own seed. Maps sharing a draw seed share the same h_1, so switching g never changes the underlying function.

1.2 What is measured

Regret. Simple regret $R_T = f_{\text{best}}(T) - f^\star$ after $T = 45$ evaluations, reported as $\log_{10} R_T$; lower is better. Because it is a log, a *gain* of $+2$ means Composite's final regret is 100 times smaller. The reference optimum $f^\star$ is computed exactly for $k = 1$: the domain is connected and h continuous, so $h([0,1]^6)$ is an interval and $\min_x g(h(x))$ is the minimum of g over that interval, which a dense grid finds for any g — including the ones a gradient method cannot descend.

$\bar{n}$. The average number of h values that could have produced the observed y,

$$\bar{n} = \mathbb{E}\big[\, |g^{-1}(y)| \,\big],$$

measured by counting crossings of $g(z) - y$ over the range h actually reaches, not inferred from the algebra. The two come apart: `shiftsq_c3` is a parabola, two-to-one on paper, but its vertex sits at $h = 3$ where h almost never goes, so in practice $\bar{n} = 1.01$.

The p-value. A one-sided paired t-test on the per-run difference, asking whether Composite's mean regret is lower. Paired because both arms share an initial design within each run, so the large variation between designs cancels. 20 runs per problem: 4 independent function draws $\times$ 5 initial designs.

The spread. Quoted as ± one standard deviation across the 20 runs, not a standard error — it describes how much a single run varies. The standard error is roughly $1/\sqrt{20}$ of it.

2 The problems

Table 1: Every problem in the grid. k is the number of intermediates and $\bar{n}$ the measured average number of h values consistent with one observation, over the range h actually reaches. A dagger marks a map excluded from the headline comparison (Table 2).

Problem	k	g	$\bar{n}$	What g discards
Bijective				
`exp_b1`	1	e^h	1.00	nothing (monotone)
`identity`	1	h	1.00	nothing
`mono3`	1	h^3	1.00	nothing (monotone, odd power)
`mono5`	1	h^5	1.00	nothing (monotone, odd power)
`root1`†	1	$h - a_1$	1.00	nothing (affine)
`shiftsq_c3`	1	$(h-3)^2$	1.01	sign of $h-3$, but h rarely exceeds 3
`sinh_b1`	1	$\sinh h$	1.00	nothing (monotone)
Not bijective				
`abs`†	1	$\lvert h \rvert$	2.00	sign of h (with a kink)
`cos_w1`	1	$\cos h$	2.03	which of many preimages
`cos_w2`	1	$\cos 2h$	5.04	which of many preimages
`cos_w4`†	1	$\cos 4h$	9.75	which of many preimages
`relu`†	1	$-\max(h, 0)$	∞	everything below zero
`root2`	1	$\prod_{i=1}^{2}(h - a_i)$	2.00	which of 2 branches
`root3`	1	$\prod_{i=1}^{3}(h - a_i)$	2.82	which of up to 3 branches
`root4`	1	$\prod_{i=1}^{4}(h - a_i)$	3.75	which of up to 4 branches
`root5`	1	$\prod_{i=1}^{5}(h - a_i)$	4.50	which of up to 5 branches
`shiftsq_c0`	1	h^2	2.00	sign of h
`shiftsq_c0.5`	1	$(h-0.5)^2$	2.00	sign of $h-0.5$
`shiftsq_c1`	1	$(h-1)^2$	1.95	sign of $h-1$
`shiftsq_c2`	1	$(h-2)^2$	1.37	sign of $h-2$
Several intermediates				
`clip_c2`†	2	$\mathrm{clip}(2h_1, -1, 1) + h_2$	–	the clipped coordinate when saturated
`sum`	2	$h_1 + h_2$	–	how the sum splits

2.1 What is excluded, and why

Four maps are kept out of the headline comparison.

- `relu`, `abs` and `clip_c2` are not everywhere differentiable — a kink or a constant piece — so they fall outside the smooth setting the argument assumes.
- `cos_w4` is deferred for separate investigation.
- `root1` is affine and therefore identical to `identity` after normalisation; it produces the same number to three decimals and is folded in.

This must be stated plainly: **the three non-smooth maps are exactly the three non-bijective maps with no effect.** Excluding them takes the non-bijective group from ten significant rows out of thirteen to ten out of ten. The criterion is defensible on its own terms, decided by smoothness of g rather than by the result, but it does flatter the grouping, so both versions are reported.

Table 2: Maps left out of the headline comparison, with the reason and what they would have contributed. The three non-smooth maps are the three non-bijective maps with no effect, so excluding them makes the grouping look cleaner than it is; both versions are reported in the text.

Problem	Why excluded	Gain	p
`relu`	kink at $h = 0$, and constant for $h < 0$	-0.154 ± 0.567	0.881
`abs`	kink at $h = 0$	$+0.098 \pm 1.000$	0.333
`clip_c2`	constant outside $[-1, 1]$	-0.003 ± 0.546	0.508
`cos_w4`	$\omega \,\mathrm{sd}(h) \approx 2.6$ rad; see text	-0.014 ± 1.691	0.515
`root1`	identical to `identity` after normalisation	$+0.087 \pm 0.693$	0.291
`sum`	linear, $k = 2$: the control	-0.230 ± 0.662	0.932

3 Results

Table 3: Bijective outer maps. Gain is Direct − Composite in final $\log_{10}$ regret, so positive favours Composite; p is a one-sided paired t-test of whether that gain exceeds zero, over the 20 runs. Only `shiftsq_c3` shows an effect, and Section 4 explains why it is not a counterexample.

Problem	$\bar{n}$	Direct	Composite	Gain	p
`shiftsq_c3`	1.01	-4.694 ± 1.611	-7.053 ± 2.174	$+\mathbf{2.359} \pm 2.053$	<0.001
`exp_b1`	1.00	-2.594 ± 0.425	-2.742 ± 0.627	$+\mathbf{0.148} \pm 0.519$	0.109
`identity`	1.00	-1.027 ± 0.793	-1.114 ± 0.606	$+\mathbf{0.087} \pm 0.693$	0.291
`mono3`	1.00	$+0.093 \pm 0.491$	$+0.012 \pm 0.445$	$+\mathbf{0.081} \pm 0.494$	0.235
`sinh_b1`	1.00	$+0.245 \pm 0.503$	$+0.164 \pm 0.401$	$+\mathbf{0.081} \pm 0.458$	0.220
`mono5`	1.00	$+0.745 \pm 0.687$	$+0.793 \pm 0.356$	$-\mathbf{0.048} \pm 0.701$	0.618

Table 4: Non-bijective outer maps, same convention. Every row is significant.

Problem	$\bar{n}$	Direct	Composite	Gain	p
`root2`	2.00	-4.975 ± 1.266	-8.395 ± 1.341	$+\mathbf{3.420} \pm 1.826$	<0.001
`shiftsq_c1`	1.95	-5.357 ± 1.055	-8.218 ± 1.054	$+\mathbf{2.861} \pm 1.337$	<0.001
`shiftsq_c0`	2.00	-5.140 ± 1.116	-7.949 ± 1.180	$+\mathbf{2.809} \pm 1.805$	<0.001
`shiftsq_c2`	1.37	-5.620 ± 0.991	-8.400 ± 1.255	$+\mathbf{2.780} \pm 1.558$	<0.001
`shiftsq_c0.5`	2.00	-5.398 ± 1.237	-7.811 ± 0.639	$+\mathbf{2.413} \pm 1.102$	<0.001
`root4`	3.75	-4.805 ± 0.943	-7.131 ± 0.781	$+\mathbf{2.326} \pm 0.915$	<0.001
`cos_w1`	2.03	-4.361 ± 1.813	-5.794 ± 1.924	$+\mathbf{1.433} \pm 2.135$	0.004
`cos_w2`	5.04	-4.938 ± 1.449	-5.997 ± 0.911	$+\mathbf{1.059} \pm 1.575$	0.004
`root5`	4.50	$+1.691 \pm 0.197$	$+1.266 \pm 0.738$	$+\mathbf{0.424} \pm 0.636$	0.004
`root3`	2.82	$+0.541 \pm 0.403$	$+0.279 \pm 0.495$	$+\mathbf{0.262} \pm 0.584$	0.030

Every non-bijective map gains; one bijective map does. The mean gain is +1.98 for the non-bijective group and +0.45 for the bijective group, and that +0.45 is almost entirely `shiftsq_c3`: without it the bijective mean is +0.07, which is nothing.

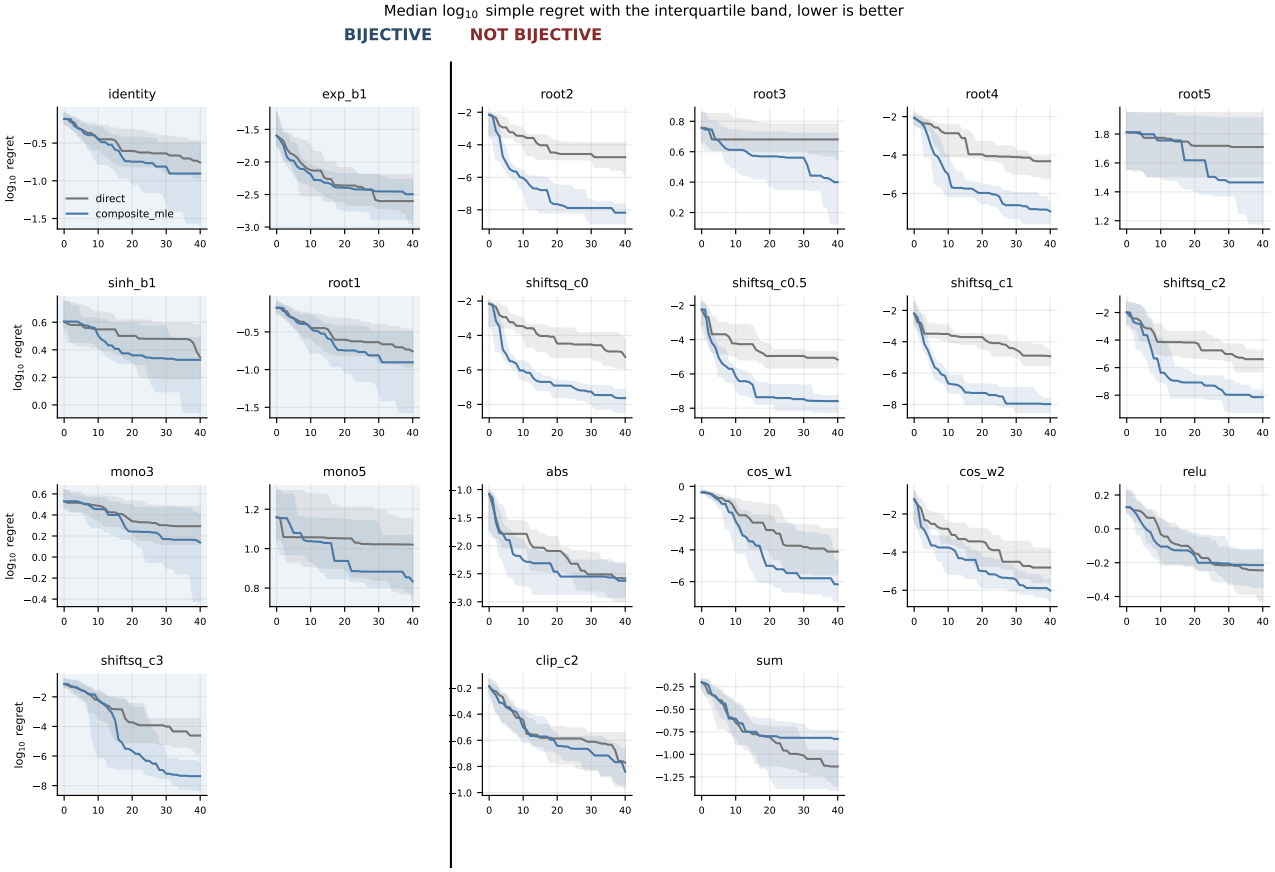


Figure 1: Median $\log_{10}$ regret against evaluation, interquartile band shaded. Bijective maps left of the rule, non-bijective right. The two blocks now look different, which is the change from the earlier grid: on the left the curves lie on top of each other, on the right Composite separates within the first ten evaluations. `shiftsq_c3`, bottom left, is the exception that Section 4 explains.

3.1 The linear control

`sum`, $g(h_1, h_2) = h_1 + h_2$, is the decisive row. Observing $y = h_1 + h_2$ tells you the sum but not the split, so information genuinely *is* destroyed — and yet the gain is -0.230, in Direct's favour.

The reason is that a linear combination of Gaussian processes is a Gaussian process. Direct's model is therefore correctly specified, and it loses nothing by not seeing the pieces. **Information loss with no misspecification buys Composite nothing.** Whatever drives the gain elsewhere, it is not the mere fact that g is many-to-one.

4 Why bijectivity predicts the gain without being the cause

Two observations do not fit an information story.

$\bar{n}$ does not order the gain. Across the single-intermediate maps the correlation between $\bar{n}$ and the gain is $r = -0.12$ — no relationship at all. `cos_w2` has $\bar{n} = 5.04$ and gains $+1.06$; `root2` has $\bar{n} = 2.00$ and gains $+3.42$.

`shiftsq_c3` gains without being many-to-one. It measures $\bar{n} = 1.01$, so a typical observation determines h, and it still gains $+2.36$ — the largest gain of any map outside the `root/shiftsq` families.

4.1 What does predict it

The quantity that works is how flat g is *at its own minimum*. Measure the plateau width: the fraction of h's reachable range mapping to within 10^{-6} of $\min g$.

Table 5: Flatness of g at its own optimum, against the measured gain. The width is the fraction of h's reachable range mapping to within 10^{-6} of $\min g$; 10^{-6} is the grid resolution, so that value means no plateau at all. Across these maps the correlation with the gain is $r = +0.78$, against $r = -0.12$ for $\bar{n}$.

Problem	Bijective?	Plateau width	Gain
`root4`	no	3.5×10^{-3}	$+2.326$
`shiftsq_c3`	yes	1.7×10^{-3}	$+2.359$
`shiftsq_c2`	no	1.5×10^{-3}	$+2.780$
`shiftsq_c1`	no	1.2×10^{-3}	$+2.861$
`shiftsq_c0.5`	no	1.1×10^{-3}	$+2.413$
`root2`	no	1.0×10^{-3}	$+3.420$
`shiftsq_c0`	no	1.0×10^{-3}	$+2.809$
`cos_w1`	no	9.7×10^{-4}	$+1.433$
`cos_w2`	no	4.8×10^{-4}	$+1.059$
`exp_b1`	yes	4.6×10^{-4}	$+0.148$
`identity`	yes	1.5×10^{-6}	$+0.087$
`mono3`	yes	5.0×10^{-7}	$+0.081$
`mono5`	yes	5.0×10^{-7}	-0.048
`root3`	no	5.0×10^{-7}	$+0.262$
`root5`	no	5.0×10^{-7}	$+0.424$
`sinh_b1`	yes	5.0×10^{-7}	$+0.081$

The table separates into two blocks with nothing in between. Above the rule every map has a genuine plateau and the mean gain is $+2.05$; below it the width is at the grid resolution, meaning the minimum is a single point, and the mean gain is $+0.10$. The correlation is $r = +0.78$, against $r = -0.12$ for $\bar{n}$.

4.2 The mechanism

Composite models h accurately. Near a flat optimum a small error in h produces a very small error in f, so an accurate model of h converts into enormous depth in objective space. Direct, looking at f alone, sees a nearly constant region and cannot resolve within it. Where g is steep at its optimum there is no such leverage and both arms do the same thing.

This explains why bijectivity predicts the gain anyway. A bijective g on an interval is monotone; a monotone function attains its minimum at an endpoint of the range; and an endpoint is generically a steep place, not a flat one. So

$$\text{bijective} \Rightarrow \text{monotone} \Rightarrow \text{optimum at an endpoint} \Rightarrow \text{steep there} \Rightarrow \text{no plateau} \Rightarrow \text{no gain.}$$

Bijectivity forecloses a flat optimum. That is why the prediction holds, and it is not an information argument at any step.

`shiftsq_c3` is the case that distinguishes the two explanations. Its fold sits at $h = 3$, at the very edge of where h reaches, so almost every observation lands on the monotone left branch and $\bar{n} \approx 1$. But its *minimum* is exactly at that fold, where g is flat. $\bar{n}$ averages over the whole distribution of y,

which is dominated by regions far from the optimum, so it misses the one place that matters. The plateau measure looks only at the optimum, and gets it right.

5 Regression error and optimisation gain come apart

The clearest evidence that this is not about model quality on average: fit both models by plain regression, with no optimiser anywhere, and compare the held-out error with the BO gain on the same problem.

Table 6: Held-out regression error against the optimisation gain, on the same problems. No acquisition function and no optimiser anywhere: 80 training points, 20 held out, 60 fits per map. Each RMSE is divided by the spread of whatever it is predicting, so 1.0 means no better than predicting the mean. Both arms fit their own hyperparameters. Produced by `python three_arm/misspec_demo.py --kernel matern --no-oracle`.

	Direct	Composite		BO gain
Problem	on f	on h (inner)	on f (outer)	(Table 3–4)
`identity`	0.786	0.786	0.786	+0.087
`mono3`	1.110	0.786	**0.976**	+0.081
`mono5`	1.996	0.786	**1.570**	−0.048
`root3`	1.253	0.786	**1.086**	+0.262
`shiftsq_c0`	1.106	0.786	**0.996**	+2.809
`shiftsq_c3`	0.794	0.786	0.782	+2.359

Three things to read off it.

The inner RMSE is 0.786 **in every row.** It must be: h is literally the same function in all of them, and Composite always fits the same GP to it. A check that nothing else is varying.

`mono3` **shows real misspecification and no gain.** Direct scores 1.110 — worse than predicting the mean of f — while Composite scores 0.976 on the same eighty points with its own fitted hyperparameters. Nothing is hidden from Direct: h^3 is bijective, so y determines h exactly. The gap is purely that $f = g(h)$ is *not a Gaussian process* when g is nonlinear, so Direct is fitting the wrong shape of model. And the BO gain is +0.081, which is nothing. **Misspecification is real here and does not convert into optimisation performance.**

`shiftsq_c3` **shows no misspecification and a huge gain.** 0.794 against 0.782 — the two models are equally good at predicting f on average — and the BO gain is +2.36. `mono5` makes the same point from the other side: the largest regression gap in the table, 1.996 against 1.570, and a BO gain of −0.048.

So average predictive accuracy and optimisation gain are decoupled. What matters is resolution *at the optimum*, which is what the plateau measures and what neither $\bar{n}$ nor RMSE captures. `identity` is the control throughout: 0.786 against 0.786, identical to three figures, because there $f = h$ and the two models are the same model.

6 What this means

- **The proposition's prediction holds.** Bijective g: no gain, in five of six cases. Non-bijective g: a gain every time, up to 3.4 orders of magnitude.
- **Its stated reason does not.** $\bar{n}$ is uncorrelated with the size of the gain, `sum` destroys information and gains nothing, and `shiftsq_c3` preserves it and gains a great deal.
- **The operative property is flatness of g at its optimum.** Bijectivity predicts the gain because it rules that out, not because of anything to do with information.
- **The earlier three-channel decomposition is withdrawn.** The split into hyperparameter estimation, model shape and information was measured on a grid where Composite held the true kernel; with both arms fitting, there is no bijective-case gain left to decompose.

Still open. `cos_w4` has the most preimages of any map ($\bar{n} = 9.75$) and a plateau, yet gains nothing (-0.014). The likely reason is that at $\omega = 4$ the posterior uncertainty in h spans most of a half-period, so Composite's predictive on f becomes as wide as f itself and knowing h stops telling it anything about the objective. That is a budget question — $\mathrm{sd}(h \mid \text{data})$ shrinks as data arrives — and is being investigated separately.

Reproducibility. All numbers come from `three_arm/results_matern2` (22 maps $\times$ 4 draws $\times$ 5 seeds). Every table in this document is generated by `python three_arm/tables.py`, so it cannot drift from the stored results. The grid was verified before use: $f^\star$ recomputes to 10^{-9} on all 88 conditions, h is shared across maps at each draw, no run has negative regret, and none sits at the regret floor.